Lecture2Quiz SEA

From classroom audio to bilingual study packs

Valsea Speech AI | AWS Bedrock | FastAPI | React + Tailwind

Press → or Space to navigate | Esc for overview | ? for help

The Problem

- Students in Southeast Asia attend lectures in **mixed languages** (English + local)
- Taking notes manually is **slow and incomplete**
- No easy way to generate **quizzes or flashcards** from a lecture
- Existing tools don't understand **SEA accents & languages** (Singlish, Vietnamese, Thai, etc.)

Our Solution

Upload a lecture recording → Get a complete bilingual study pack:

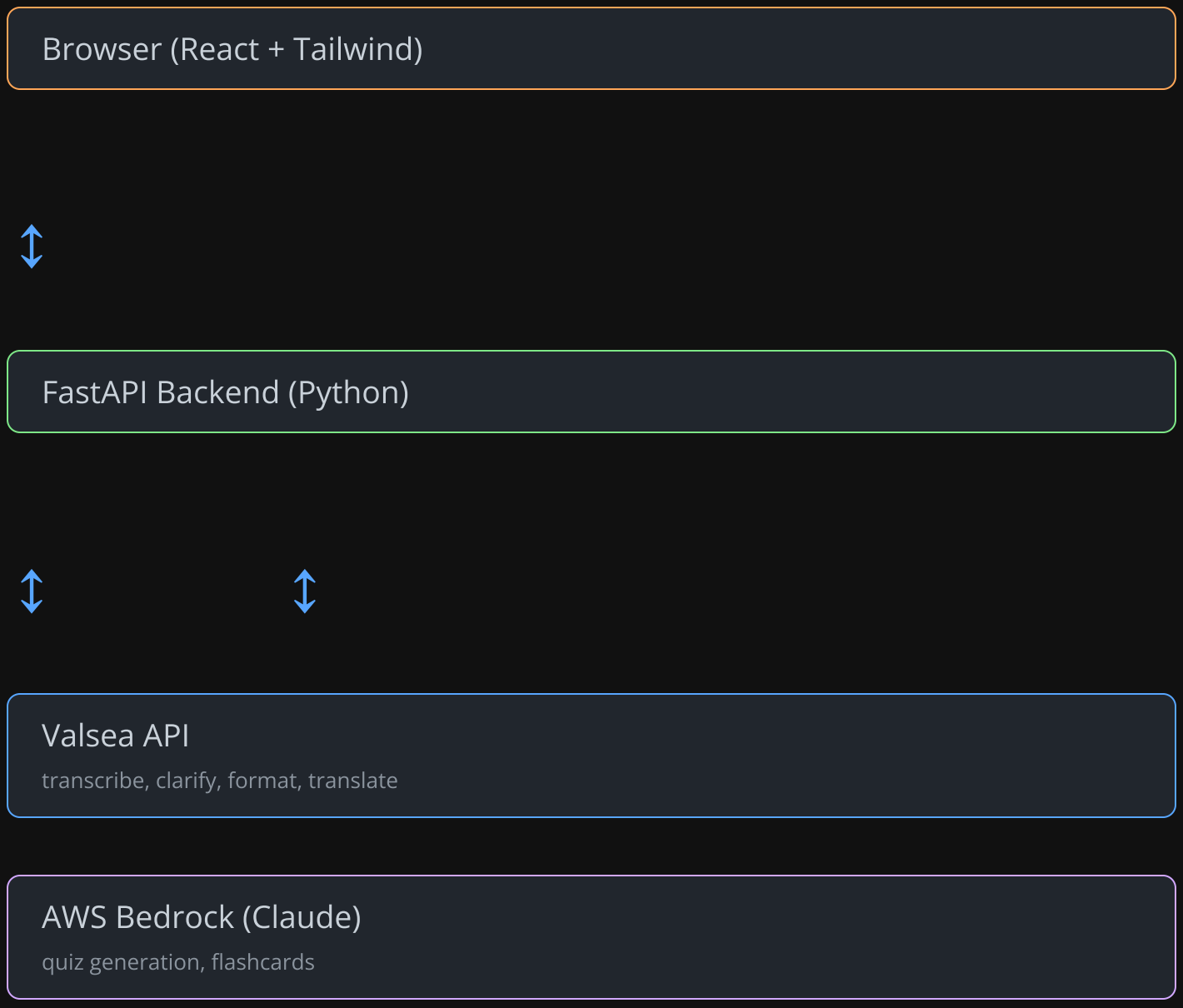
Outputs

- Clean transcript
- Key quotes & overview
- Action items / takeaways
- Bilingual glossary
- 10-question quiz
- Leveled flashcards (easy / medium / hard)

Key Features

- Supports 70+ languages
- Auto-splits large audio
- Parallel processing
- Real-time SSE progress
- Spaced-repetition study
- Quiz miss → flashcard

Architecture Overview



Processing Pipeline

1. Upload Audio

→

2. Auto-Split
(if > 8 MB)

→

3. Transcribe
Valsea STT

↓

4. Clarify
clean up noisy text

→

5. Format + Quiz
parallel (Valsea + Bedrock)

→

6. Translate
→ target language

Steps 5 runs formatting (key quotes, overview, takeaways), quiz gen, and flashcard gen **all in parallel**.

Valsea APIs Used

Endpoint	Model	Purpose
POST /v1/audio/transcriptions	valsea-transcribe	Speech-to-text with SEA accent support
POST /v1/clarifications	valsea-clarify	Clean noisy/colloquial transcript
POST /v1/formatting	valsea-format	Generate key_quotes, meeting_minutes, action_items
POST /v1/translations	valsea-translate	Translate to student's native language

All powered by **valsea.ai** — built specifically for Southeast Asian languages.

AWS Bedrock (Claude)

Used for **intelligent content generation** from the clean transcript:

Quiz Generation

- 10 multiple-choice questions
- 4 options each (A–D)
- Correct answer + explanation
- Covers key concepts from lecture

Flashcard Generation

- Front (question) / Back (answer)
- 3 difficulty levels: easy, medium, hard
- Card types: definition, concept, application
- Feeds into spaced-repetition engine

Frontend Experience

Page	Route	What it does
Home	/	Upload audio, select languages, see real-time progress (SSE)
Library	/library	Browse past sessions, view bilingual summaries & glossary
Quiz	/quiz	Interactive quiz room — immediate feedback, score tracking
Flashcards	/flashcards	Flip cards with spaced-repetition, personalized from quiz misses

Smart Audio Handling

Large lecture recordings (> 8 MB) are handled automatically:

1. Detect file exceeds threshold → trigger split
2. Split into ~4.5-minute MP3 chunks via `ffmpeg`
3. Transcribe all chunks **in parallel** (3 concurrent by default)
4. Recombine text + semantic tags in order
5. Continue pipeline with full combined transcript

No manual splitting needed — just upload the full recording (up to 1 hour).

Real-Time Progress (SSE)

The UI shows live pipeline status via Server-Sent Events:

```
POST /process?stream=true

← {"type":"phase","phase":"transcribe","label":"Transcribing audio…"}
← {"type":"phase","phase":"clarify","label":"Clarifying transcript…"}
← {"type":"phase","phase":"summary_quiz","label":"Generating study pack…"}
← {"type":"phase","phase":"translate","label":"Translating → vietnamese…"}
← {"type":"complete","payload":{ transcript, summary, quiz, flashcards }}
```

Students see exactly what's happening — no mystery loading spinner.

Student Study Flow

Upload Lecture

→

Read Summary
(bilingual)

→

Take Quiz

↓

Review Flashcards
(includes quiz misses!)

→

Spaced Repetition
(localStorage)

Wrong quiz answers automatically become flashcards for extra review.

Tech Stack

Backend

- Python + FastAPI
- httpx (async HTTP)
- boto3 (AWS Bedrock)
- ffmpeg (audio splitting)
- pytest + respx (testing)

Frontend

- React + React Router
- Vite (build tool)
- Tailwind CSS
- localStorage (persistence)
- EventSource (SSE)

API Design

Method	Path	Description
GET	/health	Liveness check
POST	/process	Main pipeline endpoint multipart: `audio`, `target_language`, `transcription_language`, `stream` Returns: transcript, summary (bilingual), quiz, flashcards

Single endpoint does everything — upload once, get the complete study pack.

Quick Demo (Mock Mode)

Run without API keys using mock data:

```
# .env
USE_MOCK_QUIZ=true
USE_MOCK_FLASHCARDS=true
```

This loads pre-built JSON fixtures so you can demo the UI without:

- Valsea API key
- AWS credentials
- Network connectivity

Language Support

Valsea supports **70+ languages** with accent-aware models:

English Vietnamese Thai Indonesian Malay Filipino Chinese Japanese

Korean Singlish Hindi Khmer Lao Javanese Tamil Bengali

+ 55 more...

Including regional variants: English-PH, Arabic-UAE, Bengali-BD, etc.

Running the Project

```
# Terminal 1 — Backend
cd backend
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
uvicorn main:app --reload --port 8001

# Terminal 2 — Frontend
cd frontend
npm install && npm run dev
```

Open http://127.0.0.1:5173 → Upload audio → Watch the magic happen.

What Makes This Unique

1. **SEA-first** — built on Valsea, which understands Singlish, accented English, and local languages
2. **End-to-end** — one upload produces transcript + summary + quiz + flashcards
3. **Bilingual output** — everything in English + student's native language
4. **Adaptive learning** — quiz mistakes feed into flashcard deck automatically
5. **No manual work** — auto-split, parallel processing, smart retries

Roadmap

- Real-time live transcription (WebSocket via Valsea RTT)
- Persistent session database (SQLite → PostgreSQL)
- Analytics dashboard (study progress, weak topics)
- Collaborative study rooms
- Mobile-friendly PWA
- Export to PDF / Anki deck

Thank You

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GitHub: VALSEA_PROJECT API: valsea.ai

Print to PDF: append `?print-pdf` to the URL, then `Ctrl+P` / `Cmd+P`

Speaker notes